

PURE 1.7.1 Differentiation

work out gradients of graphs
↳ how steep?

1. Derivatives and gradient functions

1. Power rule

"the derivate of x^n with respect to x "

$$\frac{d}{dx} (x^n) = nx^{n-1}$$

likewise,
the derivative of $f(x)$
is $f'(x)$

graph

gradient
function

e.g.: $y = x^3 \Rightarrow \frac{dy}{dx} = \underline{x^3}^{-1} = 3x^2$

$y = x^{-2} \Rightarrow \frac{dy}{dx} = \underline{x^{-2}}^{-1} = -2x^{-3}$

$y = 3x^{\frac{1}{2}} \Rightarrow \frac{dy}{dx} = 3 \times \underline{x^{\frac{1}{2}}}^{-1} = 3 \times \frac{1}{2} x^{-\frac{1}{2}} = \frac{3}{2\sqrt{x}}$

a constant

$$\frac{d}{dx} [k f(x)] = k \cdot \frac{d}{dx} [f(x)]$$

2. Scalar multiple rule

$y = 3x^4 + 2x^2 \Rightarrow \frac{dy}{dx} = 3 \cdot 4x^3 + 2 \cdot 2x^1 = 12x^3 + 4x$

3. Addition/ subtraction rule

$$\frac{d}{dx} [f(x) \pm g(x)] = \frac{d}{dx} [f(x)] \pm \frac{d}{dx} [g(x)]$$

4. The chain rule

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$

e.g.: $y = (3x-2)^7 = u^7 \Rightarrow \frac{dy}{du} = 7u^6$

$\downarrow$
 $\frac{du}{dx} = \frac{d}{dx} (3x-2) = 3$

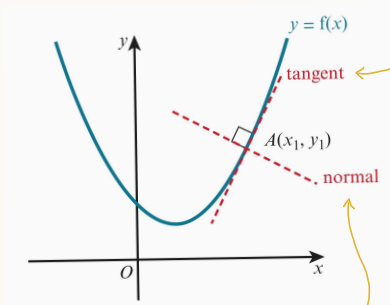
$$\frac{dy}{dx} = \frac{du}{dx} \cdot \frac{dy}{du} = 3 \times 7u^6$$

substitute
back

$$= 21u^6$$

$$= 21(3x-2)^6$$

2. Tangents and normals



$$\frac{dy}{dx}$$

gradient at point $A(x_1, y_1)$ is m

tangent

- touches curve, doesn't go through
- same gradient as curve (m)
- equation of tangent:

$$y - y_1 = m(x - x_1)$$

normal

- perpendicular $\perp$ to curve
↳ gradient: $-\frac{1}{m}$

- equation of normal:

$$y - y_1 = -\frac{1}{m}(x - x_1)$$

3. Second derivatives

- differentiate y with respect to x : $\frac{dy}{dx}$ (first derivative) → $f'(x)$
- differentiate $\frac{dy}{dx}$ with respect to x : $\frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{d^2y}{dx^2}$ (second derivative) → $f''(x)$

e.g.: $y = x^3 + 5x^2 - 3x + 2$

$$\frac{dy}{dx} = 3x^2 + 10x - 3$$

$$\frac{d^2y}{dx^2} = 6x + 10$$

or

$$f(x) = x^3 + 5x^2 - 3x + 2$$

or

$$f'(x) = 3x^2 + 10x - 3$$

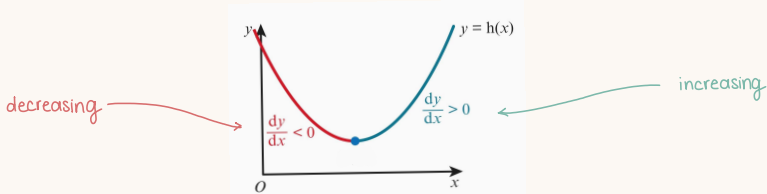
or

$$f''(x) = 6x + 10$$

Further Differentiation

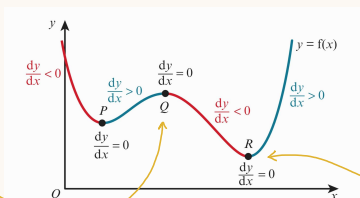
- is the function **increasing / decreasing**?
- what's its **maximum / minimum** value?
- ↳ solve practical problems involving **rates of change**.

1. Increasing / decreasing functions



2. Stationary points

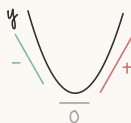
$$\frac{dy}{dx} = 0$$



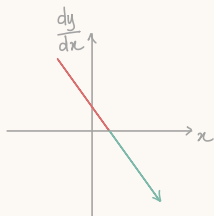
1. maximum point



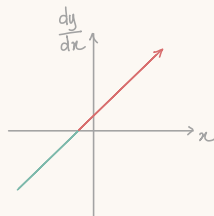
2. minimum point



← curve y →



← gradient of curve y $\frac{dy}{dx}$ →



gradient of $\frac{dy}{dx}$ is **negative**

← second derivative →

gradient of $\frac{dy}{dx}$ is **positive**

$$\text{if } \frac{dy}{dx} = 0 \text{ and } \frac{d^2y}{dx^2} < 0$$

→ maximum point

$$\frac{d^2y}{dx^2}$$

$$\text{if } \frac{dy}{dx} = 0 \text{ and } \frac{d^2y}{dx^2} > 0$$

→ minimum point

3. Rates of change

a.k.a. a change over time

e.g.: $\frac{\text{distance}}{\text{time}} = \text{speed}$

$\frac{\text{speed}}{\text{time}} = \text{acceleration}$

The chain rule states:

$$\frac{dy}{dt} = \frac{dy}{dx} \times \frac{dx}{dt}$$

$$\frac{dx}{dy} = \frac{1}{\frac{dy}{dx}}$$

← $\frac{1}{\text{reciprocal}}$

✧ practical applications

- rate of increase in size of an oil leak
- the change in surface area and volume when you blow up a balloon
- rate at which water drains from a container